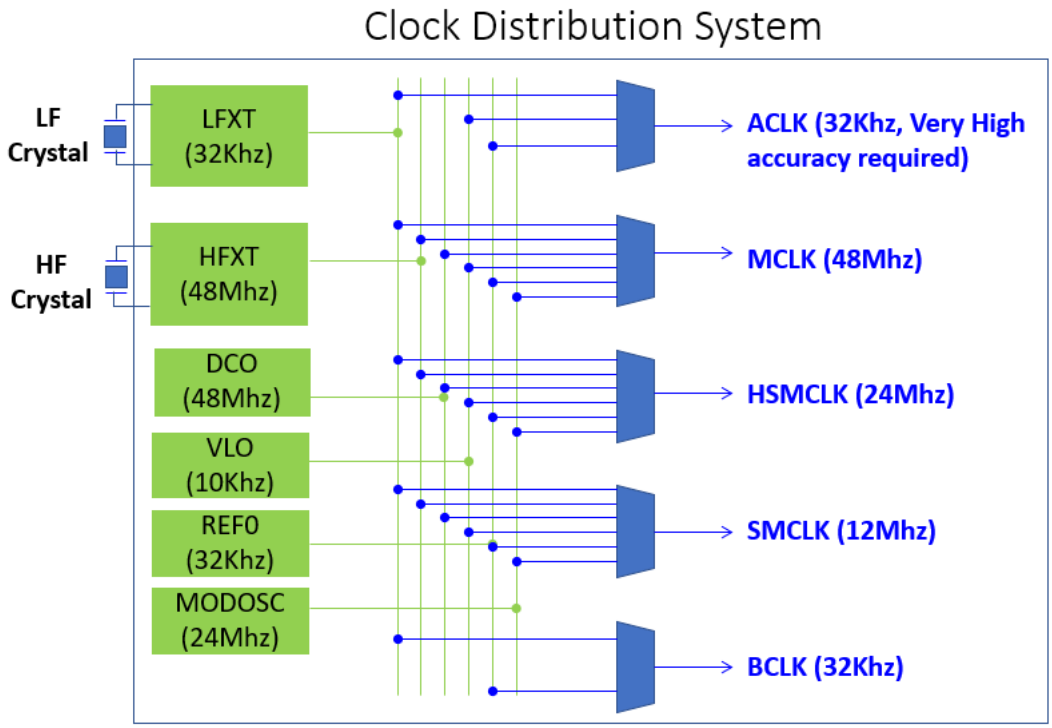
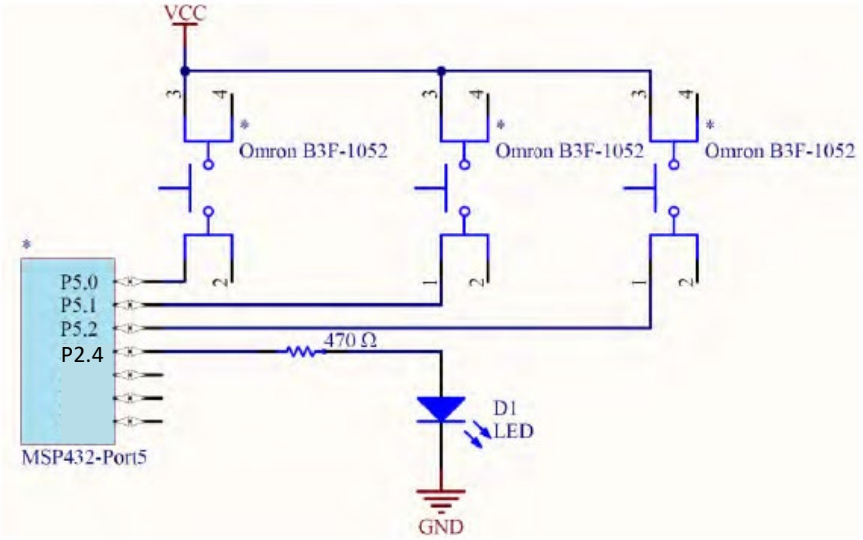
Clock Tree and Timers

1. Diagram below shows the clock distribution system of a MSP432 processor system, illustrating the valid connection between clock sources and the internal clock rails. Explain which clock source (e.g. DCO) can be used to clock the clock rails (e.g. MCLK). Note the following points.
   - Not all clock sources need to be used.
   - Priority is cost followed by power consumption, but all functional requirement needs to be met.
   - Detail operations of each clock source module is not needed, other than points mentioned in this question.
   - MODOSC is not available when MSP432 is in power down mode.
   - SMCLK is required when MSP432 is in power down mode.
   - Assume there are clock dividers to step-down the selected clock source frequency to the required clock rail frequency.



2. The diagram below shows the schematic of an Intrusion Alarm System that has the following specifications.

- Pin 2 and 4 of the Omron B3F-1052 is not connected to anywhere.
- If the Omron switches are opened, it implies that the windows are opened.
- MSP432 Port pins use interrupt mechanism to detect any change in the contact status of the switch
- If any windows are opened, the LED will blink at a rate of 5Hz, else, it'll remain OFF. Timer_A0 is used to generate a 5Hz PWM signal at 50% duty cycle for the LED blinking. (Yes, I know… not much of an alarm …).



a. Perform the necessary registers configuration for the GPIO and Timer peripherals

- GPIO
  - P5.0, P5.1 and P5.2 port pins initialization
  - Interrupt configuration for Port 5.
  - Set priority of the interrupt to 2.
  - Only need to initialize the following registers
    - For GPIO: PxOUT, PxDIR, PxREN, PxSEL0, PxSEL1, PxIES, PxIE and PxIFG.
    - For interrupt: ISER1, ICPR1, IPR9. Why did we choose these registers for Port5?

b. Timer
  - Timer_A0 in compare mode to generate the 5Hz PWM signal out from P2.4 pin.

- Assume the clock source of Timer_A0 is SMCLK and SMCLK is 1Mhz. Use the clock dividers and period registers in Timer_A0 to derive a 5Hz signal.
- Only need to initialize the following registers
  - GPIO: P2DIR, P2SEL0, P2SEL1
  - Timer: TA0CTL, TA0CCTLx, TA0CCRx, TA0EX0

The necessary documents have been extracted for you and can be found in the appendix of the tutorial.

Appendix

GPIO

6.12.10 Port P5 (P5.0 to P5.5) Input/Output With Schmitt Trigger

Figure 6-9 shows the port diagram. Table 6-71 summarizes the selection of the pin functions.

Table 6-71. Port P5 (P5.0 to P5.5) Pin Functions

PIN NAME (P5.x)	x	FUNCTION	CONTROL BITS OR SIGNALS(1)		
			P5DIR.x	P5SEL1.x	P5SEL0.x
P5.0/A5	0	P5.0 (I/O)	I: 0; O: 1	0	0
		N/A	0	0	1
		DVSS	1		
		N/A	0	1	0
		DVSS	1		
		A5(2)	X	1	1
P5.1/A4	1	P5.1 (I/O)	I: 0; O: 1	0	0
		N/A	0	0	1
		DVSS	1		
		N/A	0	1	0
		DVSS	1		
		A4(2)	X	1	1
P5.2/A3	2	P5.2 (I/O)	I: 0; O: 1	0	0
		N/A	0	0	1
		DVSS	1		
		N/A	0	1	0
		DVSS	1		
		A3(2)	X	1	1

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

7	6	5	4	3	2	1	0
PxOUT							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

12.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

12.4.5 PxREN Register

Port X Pullup or Pulldown Resistor Enable Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-5. PxREN Register

7	6	5	4	3	2	1	0
PxREN							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-8. PxREN Register Description

Bit	Field	Type	Reset	Description
7-0	PxREN	RW	0h	Port X pullup or pulldown resistor enable. When the port is configured as an input, setting this bit enables or disables the pullup or pulldown. 0b = Pullup or pulldown disabled 1b = Pullup or pulldown enabled

12.4.7 PxSEL0 Register

Port X Function Selection Register 0 (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-7. PxSEL0 Register

7	6	5	4	3	2	1	0
PxSEL0							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-10. PxSEL0 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL0	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. See PxSEL1 for the definition of each value.

12.4.8 PxSEL1 Register

Port X Function Selection Register 1 (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-8. PxSEL1 Register

7	6	5	4	3	2	1	0
PxSEL1							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-11. PxSEL1 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL1	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. 00b = General-purpose I/O is selected 01b = Primary module function is selected 10b = Secondary module function is selected 11b = Tertiary module function is selected

12.4.10 PxIES Register

Port X Interrupt Edge Select Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10)

Figure 12-10. PxIES Register

7	6	5	4	3	2	1	0
PxIES							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-13. P1IES Register Description

Bit	Field	Type	Reset	Description
7-0	PxIES	RW	Undefined	Port X interrupt edge select 0b = PxIFG flag is set with a low-to-high transition. 1b = PxIFG flag is set with a high-to-low transition.

12.4.11 PxIE Register

Port X Interrupt Enable Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10)

Figure 12-11. PxIE Register

7	6	5	4	3	2	1	0
PxIE							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-14. PxIE Register Description

Bit	Field	Type	Reset	Description
7-0	PxIE	RW	0h	Port X interrupt enable 0b = Corresponding port interrupt disabled 1b = Corresponding port interrupt enabled

12.4.12 PxIFG Register

Port X Interrupt Flag Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10)

Figure 12-12. PxIFG Register

7	6	5	4	3	2	1	0
PxIFG							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-15. PxIFG Register Description

Bit	Field	Type	Reset	Description
7-0	PxIFG	RW	0h	Port X interrupt flag 0b = No interrupt is pending. 1b = Interrupt is pending.

NVIC

NVIC Table (P118 MSP432 Datasheets)

Table 6-39. NVIC Interrupts (continued)

NVIC INTERRUPT INPUT	SOURCE	FLAGS IN SOURCE
INTISR[4]	FPU_INT (2)	Combined interrupt from flags in the FPSCR (part of Cortex-M4 FPU)
INTISR[5]	FLCTL	Flash Controller interrupt flags
INTISR[6]	COMP_E0	Comparator_E0 interrupt flags
INTISR[7]	COMP_E1	Comparator_E1 interrupt flags
INTISR[8]	Timer_A0	TA0CCTL0.CCIFG
INTISR[9]	Timer_A0	TA0CCTLx.CCIFG (x = 1 to 4), TA0CTL.TAIFG
INTISR[10]	Timer_A1	TA1CCTL0.CCIFG
INTISR[11]	Timer_A1	TA1CCTLx.CCIFG (x = 1 to 4), TA1CTL.TAIFG
INTISR[12]	Timer_A2	TA2CCTL0.CCIFG
INTISR[13]	Timer_A2	TA2CCTLx.CCIFG (x = 1 to 4), TA2CTL.TAIFG
INTISR[14]	Timer_A3	TA3CCTL0.CCIFG
INTISR[15]	Timer_A3	TA3CCTLx.CCIFG (x = 1 to 4), TA3CTL.TAIFG
INTISR[16]	eUSCI_A0	UART or SPI mode TX, RX, and Status Flags
INTISR[17]	eUSCI_A1	UART or SPI mode TX, RX, and Status Flags
INTISR[18]	eUSCI_A2	UART or SPI mode TX, RX, and Status Flags
INTISR[19]	eUSCI_A3	UART or SPI mode TX, RX, and Status Flags
INTISR[20]	eUSCI_B0	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[21]	eUSCI_B1	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[22]	eUSCI_B2	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[23]	eUSCI_B3	SPI or I^2C mode TX, RX, and Status Flags (I^2C in multiple-slave mode)
INTISR[24]	Precision ADC	IFG[0-31], LO/IN/HI-IFG, RDYIFG, OVIFG, TOVIFG
INTISR[25]	Timer32_INT1	Timer32 interrupt for Timer1
INTISR[26]	Timer32_INT2	Timer32 interrupt for Timer2
INTISR[27]	Timer32_INTC	Timer32 Combined interrupt
INTISR[28]	AES256	AESRDYIFG
INTISR[29]	RTC_C	OFIFG, RDYIFG, TEVIFG, AIFG, RT0PSIFG, RT1PSIFG
INTISR[30]	DMA_ERR	DMA error interrupt
INTISR[31]	DMA_INT3	DMA completion interrupt3
INTISR[32]	DMA_INT2	DMA completion interrupt2
INTISR[33]	DMA_INT1	DMA completion interrupt1
INTISR[34]	DMA_INT0 (3)	DMA completion interrupt0
INTISR[35]	I/O Port P1	P1IFG.x (x = 0 to 7)
INTISR[36]	I/O Port P2	P2IFG.x (x = 0 to 7)
INTISR[37]	I/O Port P3	P3IFG.x (x = 0 to 7)
INTISR[38]	I/O Port P4	P4IFG.x (x = 0 to 7)
INTISR[39]	I/O Port P5	P5IFG.x (x = 0 to 7)
INTISR[40]	I/O Port P6	P6IFG.x (x = 0 to 7)
INTISR[41]	Reserved	
INTISR[42]	Reserved	

2.4.3.2 ISER1 Register (Offset = 104h) [reset = 00000000h]

ISER1 is shown in Figure 2-21 and described in Table 2-27.

Irq 32 to 63 Set Enable Register. Use the Interrupt Set-Enable Registers to enable interrupts and determine which interrupts are currently enabled.

Figure 2-21. ISER1 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SETENA																															
R/W-0h																															

Table 2-27. ISER1 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	SETENA	R/W	0h	Writing 0 to a SETENA bit has no effect, writing 1 to a bit enables the corresponding interrupt. Reading the bit returns its current enable state. Reset clears the SETENA fields.

2.4.3.8 ICPR1 Register (Offset = 284h) [reset = 00000000h]

ICPR1 is shown in Figure 2-27 and described in Table 2-33.

Irq 32 to 63 Clear Pending Register. Use the Interrupt Clear-Pending Registers to clear pending interrupts and determine which interrupts are currently pending.

Figure 2-27. ICPR1 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CLRPEND																															
R/W-0h																															

Table 2-33. ICPR1 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	CLRPEND	R/W	0h	Writing 0 to a CLRPEND bit has no effect, writing 1 to a bit clears the corresponding pending interrupt. Reading the bit returns its current state.

2.4.3.20 IPR9 Register (Offset = 424h) [reset = 00000000h]

IPR9 is shown in Figure 2-39 and described in Table 2-45.

Irq 36 to 39 Priority Register. Use the Interrupt Priority Registers to assign a priority from 0 to 255 to each of the available interrupts. 0 is the highest priority, and 255 is the lowest.

The hardware priority mechanism only looks at the upper N bits of the priority field (where N is 3 for the MSP432 family), so any prioritization must be performed in those bits. The remaining bits can be used to sub-prioritize the interrupt sources, and may be used by the hardware priority mechanism on a future part

Figure 2-39. IPR9 Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRI_39								PRI_38								PRI_37								PRI_36							
R/W-0h								R/W-0h								R/W-0h								R/W-0h							

Table 2-45. IPR9 Register Field Descriptions

Bit	Field	Type	Reset	Description
31-24	PRI_39	R/W	0h	Priority of interrupt 39
23-16	PRI_38	R/W	0h	Priority of interrupt 38
15-8	PRI_37	R/W	0h	Priority of interrupt 37
7-0	PRI_36	R/W	0h	Priority of interrupt 36

Timer

Table 6-67. Port P2 (P2.4 to P2.7) Pin Functions

PIN NAME (P2.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]			
			P2DIR.x	P2SEL1.x	P2SEL0.x	P2MAPx
P2.4/PM_TA0.1[2]	4	P2.4 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI1A	0	0	1	default
		TA0.1	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.5/PM_TA0.2[2]	5	P2.5 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI2A	0	0	1	default
		TA0.2	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			

19.3.1 TAxCTL Register

Timer_Ax Control Register

Figure 19-15. TAxCTL Register

15	14	13	12	11	10	9	8
Reserved						TASSEL	
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0
7	6	5	4	3	2	1	0
ID		MC		Reserved	TACLR	TAIE	TAIFG
rw-0	rw-0	rw-0	rw-0	rw-0	w-0	rw-0	rw-0

Table 19-4. TAxCTL Register Description

Bit	Field	Type	Reset	Description
15-10	Reserved	RW	0h	Reserved
9-8	TASSEL	RW	0h	Timer_A clock source select 00b = TAxCLK 01b = ACLK 10b = SMCLK 11b = INCLK
7-6	ID	RW	0h	Input divider. These bits along with the TAIDEX bits select the divider for the input clock. 00b = /1 01b = /2 10b = /4 11b = /8
5-4	MC	RW	0h	Mode control. Setting MCx = 00h when Timer_A is not in use conserves power. 00b = Stop mode: Timer is halted 01b = Up mode: Timer counts up to TAxCCR0 10b = Continuous mode: Timer counts up to 0FFFFh 11b = Up/down mode: Timer counts up to TAxCCR0 then down to 0000h
3	Reserved	RW	0h	Reserved
2	TACLR	RW	0h	Timer_A clear. Setting this bit resets TAxR, the timer clock divider logic, and the count direction. The TACLR bit is automatically reset and is always read as zero.
1	TAIE	RW	0h	Timer_A interrupt enable. This bit enables the TAIFG interrupt request. 0b = Interrupt disabled 1b = Interrupt enabled
0	TAIFG	RW	0h	Timer_A interrupt flag 0b = No interrupt pending 1b = Interrupt pending

19.3.3 TAxCCTL0 to TAxCCTL6 Register

Timer_Ax Capture/Compare Control 0 Register to Timer_Ax Capture/Compare Control 6

Figure 19-17. TAxCCTL0 to TAxCCTL6 Register

15	14	13	12	11	10	9	8
CM		CCIS		SCS	SCCI	Reserved	CAP
rw-0	rw-0	rw-0	rw-0	rw-0	r-0	r-0	rw-0
7	6	5	4	3	2	1	0
OUTMOD			CCIE	CCI	OUT	COV	CCIFG
rw-0	rw-0	rw-0	rw-0	r	rw-0	rw-0	rw-0

Table 19-6. TAxCCTL0 to TAxCCTL6 Register Description

Bit	Field	Type	Reset	Description
15-14	CM	RW	0h	Capture mode 00b = No capture 01b = Capture on rising edge 10b = Capture on falling edge 11b = Capture on both rising and falling edges
13-12	CCIS	RW	0h	Capture/compare input select. These bits select the TAxCCR0 input signal. See the device-specific data sheet for specific signal connections. 00b = CCIxA 01b = CCIxB 10b = GND 11b = VCC
11	SCS	RW	0h	Synchronize capture source. This bit is used to synchronize the capture input signal with the timer clock. 0b = Asynchronous capture 1b = Synchronous capture
10	SCCI	RW	0h	Synchronized capture/compare input. The selected CCI input signal is latched with the EQUx signal and can be read via this bit.
9	Reserved	R	0h	Reserved. Reads as 0.
8	CAP	RW	0h	Capture mode 0b = Compare mode 1b = Capture mode
7-5	OUTMOD	RW	0h	Output mode. Modes 2, 3, 6, and 7 are not useful for TAxCCR0 because EQUx = EQU0. 000b = OUT bit value 001b = Set 010b = Toggle/reset 011b = Set/reset 100b = Toggle 101b = Reset 110b = Toggle/set 111b = Reset/set
4	CCIE	RW	0h	Capture/compare interrupt enable. This bit enables the interrupt request of the corresponding CCIFG flag. 0b = Interrupt disabled 1b = Interrupt enabled
3	CCI	R	0h	Capture/compare input. The selected input signal can be read by this bit.
2	OUT	RW	0h	Output. For output mode 0, this bit directly controls the state of the output. 0b = Output low 1b = Output high

19.3.3 TAxCCTL0 to TAxCCTL6 Register

Timer_Ax Capture/Compare Control 0 Register to Timer_Ax Capture/Compare Control 6

Figure 19-17. TAxCCTL0 to TAxCCTL6 Register

15	14	13	12	11	10	9	8
CM		CCIS		SCS	SCCI	Reserved	CAP
rw-0	rw-0	rw-0	rw-0	rw-0	r-0	r-0	rw-0
7	6	5	4	3	2	1	0
OUTMOD			CCIE	CCI	OUT	COV	CCIFG
rw-0	rw-0	rw-0	rw-0	r	rw-0	rw-0	rw-0

Table 19-6. TAxCCTL0 to TAxCCTL6 Register Description

Bit	Field	Type	Reset	Description
15-14	CM	RW	0h	Capture mode 00b = No capture 01b = Capture on rising edge 10b = Capture on falling edge 11b = Capture on both rising and falling edges
13-12	CCIS	RW	0h	Capture/compare input select. These bits select the TAxCCR0 input signal. See the device-specific data sheet for specific signal connections. 00b = CCIxA 01b = CCIxB 10b = GND 11b = VCC
11	SCS	RW	0h	Synchronize capture source. This bit is used to synchronize the capture input signal with the timer clock. 0b = Asynchronous capture 1b = Synchronous capture
10	SCCI	RW	0h	Synchronized capture/compare input. The selected CCI input signal is latched with the EQUx signal and can be read via this bit.
9	Reserved	R	0h	Reserved. Reads as 0.
8	CAP	RW	0h	Capture mode 0b = Compare mode 1b = Capture mode
7-5	OUTMOD	RW	0h	Output mode. Modes 2, 3, 6, and 7 are not useful for TAxCCR0 because EQUx = EQU0. 000b = OUT bit value 001b = Set 010b = Toggle/reset 011b = Set/reset 100b = Toggle 101b = Reset 110b = Toggle/set 111b = Reset/set
4	CCIE	RW	0h	Capture/compare interrupt enable. This bit enables the interrupt request of the corresponding CCIFG flag. 0b = Interrupt disabled 1b = Interrupt enabled
3	CCI	R	0h	Capture/compare input. The selected input signal can be read by this bit.
2	OUT	RW	0h	Output. For output mode 0, this bit directly controls the state of the output. 0b = Output low 1b = Output high

Bit	Field	Type	Reset	Description
1	COV	RW	0h	Capture overflow. This bit indicates a capture overflow occurred. COV must be reset with software. 0b = No capture overflow occurred 1b = Capture overflow occurred
0	CCIFG	RW	0h	Capture/compare interrupt flag 0b = No interrupt pending 1b = Interrupt pending

19.3.4 TAxCCR0 to TAxCCR6 Register

Timer_Ax Capture/Compare 0 Register to Timer_Ax Capture/Compare 6 Register

Figure 19-18. TAxCCR0 to TAxCCR6 Register

15	14	13	12	11	10	9	8
TAxCCRn							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0
7	6	5	4	3	2	1	0
TAxCCRn							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 19-7. TAxCCR0 to TAxCCR6 Register Description

Bit	Field	Type	Reset	Description
15-0	TAxCCR0	RW	0h	Compare mode: TAxCCRn holds the data for the comparison to the timer value in the Timer_A Register, TAxR. Capture mode: The Timer_A Register, TAxR, is copied into the TAxCCRn register when a capture is performed.

Figure 19-20. TAxEX0 Register

15	14	13	12	11	10	9	8
Reserved							
r0	r0	r0	r0	r0	r0	r0	r0
7	6	5	4	3	2	1	0
Reserved					TAIDEX(1)		
r0	r0	r0	r0	r0	rw-0	rw-0	rw-0

(1) After programming TAIDEX bits and configuration of the timer, set TACLR bit to ensure proper reset of the timer divider logic.

Table 19-9. TAxEX0 Register Description

Bit	Field	Type	Reset	Description
15-3	Reserved	R	0h	Reserved. Reads as 0.
2-0	TAIDEX	RW	0h	Input divider expansion. These bits along with the ID bits select the divider for the input clock. 000b = Divide by 1 001b = Divide by 2 010b = Divide by 3 011b = Divide by 4 100b = Divide by 5 101b = Divide by 6 110b = Divide by 7 111b = Divide by 8